

Project Design Phase-II

Solution requirements (functional and non functional)

Date	31 January 2026
Team ID	LTVIP2026TMIDS74486
Project Name	Househunt; Finding your perfect rental home
Maximum Marks	4 Marks

Functional Requirements

FR No.

Functional Requirement (Epic)

Sub Requirement (Story / Sub-Task)

FR-1

User Authentication

Card reading through ATM reader

PIN entry through keypad

Validation of card and PIN

FR-2

User Confirmation

OTP generation to registered mobile

OTP verification before transaction

FR-3

Secure Data Processing

Encryption of PIN, OTP, and transaction data using AES

Secure transmission of encrypted data to bank server

FR-4

Transaction Handling

Cash withdrawal after successful verification

Balance inquiry display

Transaction rejection and alert for invalid attempts

Non-functional Requirements

NFR No.

Non-Functional Requirement

Description

NFR-1

Usability

ATM interface should be simple, user-friendly, and easy to operate.

NFR-2

Security

Sensitive data must be protected using AES encryption and OTP verification.

NFR-3

Reliability

System should perform accurate authentication and transaction processing without errors.

NFR-4

Performance

Transactions and encryption operations must complete within a few seconds.

NFR-5

Availability

ATM service should be available 24/7 with minimal downtime.

NFR-6

Scalability

System should support increasing number of users and transactions in future.